1 I

1.1 (1)

Taking the total differential of the internal energy,

$$\begin{aligned} dU(T,V) &= \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV \\ &= C_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV. \end{aligned}$$

For a quasi-static adiabatic process, the first law of thermodynamics gives

$$dU + PdV = 0.$$

Therefore,

$$\boxed{0 = C_V dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right] dV}.$$

1.2 (2)

Expressing the internal energy as $U(T, V(T,P))$, we obtain

$$\begin{aligned} \left(\frac{\partial U(T,P)}{\partial T}\right)_P &= \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \\ &= C_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P. \end{aligned}$$

Using the definition of C_P,

$$C_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P,$$

we find

$$\boxed{C_P - C_V = \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right]\left(\frac{\partial V}{\partial T}\right)_P}.$$

1.3 (3)

Combining the results from (1) and (2),

$$0 = C_V dT + \frac{C_P - C_V}{\left(\frac{\partial V}{\partial T}\right)_P} dV.$$

Therefore, along a quasi-static adiabatic process,

$$\begin{aligned} \frac{dT}{dV} &= -\frac{C_P - C_V}{C_V} \frac{1}{\left(\frac{\partial V}{\partial T}\right)_P} \\ &= \boxed{\left(1 - \frac{C_P}{C_V}\right)\left(\frac{\partial T}{\partial V}\right)_P}. \end{aligned}$$

For an ideal gas, $PV = NkT$, so

$$\left(\frac{\partial T}{\partial V}\right)_P = \frac{T}{V}.$$

Introducing $\gamma = C_P/C_V$, we obtain

$$\frac{dT}{T} = -(\gamma - 1)\frac{dV}{V}.$$

Assuming γ is constant, integration gives

$$\boxed{TV^{\gamma-1} = \text{constant}}.$$

2 II

2.1 (4)

Since the system is an ideal gas with $U = U(T)$, the internal energy remains unchanged during an isothermal process:

$$dU = 0.$$

From the first law of thermodynamics,

$$dU = \delta Q - PdV,$$

we obtain

$$\delta Q = PdV.$$

Therefore, the heat absorbed during the isothermal expansion from V_1 to V_2 is

$$\begin{aligned} Q &= \int_{V_1}^{V_2} PdV \\ &= \int_{V_1}^{V_2} \frac{NkT_H}{V} dV \\ &= \boxed{NkT_H \ln \frac{V_2}{V_1}}. \end{aligned}$$

2.2 (5)

Since the process $2 \to 3$ is adiabatic,

$$T_H V_2^{\gamma-1} = T_L V_3^{\gamma-1}.$$

Rearranging,

$$\left(\frac{V_3}{V_2}\right)^{\gamma-1} = \frac{T_H}{T_L}.$$

Hence,

$$\boxed{V_3 = \left(\frac{T_H}{T_L}\right)^{\frac{1}{\gamma-1}} V_2}.$$

2.3 (6)

Since the ideal Carnot cycle consists of reversible processes, the entropy change is given by

$$dS = \frac{\delta Q_{\text{rev}}}{T}.$$

For the isothermal expansion $1 \to 2$,

$$\boxed{S_{1\to 2} = \frac{Q}{T_H}}.$$

For the reversible adiabatic expansion $2 \to 3$, $\delta Q_{\text{rev}} = 0$, so

$$\boxed{S_{2\to 3} = 0}.$$

Alternatively, for an ideal gas with constant C_V, the entropy change can be expressed as

$$S_{2\to 3} = C_V \ln \frac{T_L}{T_H} + Nk \ln \frac{V_3}{V_2}.$$

From the adiabatic relation,

$$\ln \frac{T_L}{T_H} = -(\gamma - 1) \ln \frac{V_3}{V_2}.$$

Using $C_P - C_V = Nk$ and $\gamma = C_P/C_V$, we have

$$C_V(\gamma - 1) = Nk.$$

Thus,

$$\begin{aligned} S_{2\to 3} &= -C_V(\gamma - 1) \ln \frac{V_3}{V_2} + Nk \ln \frac{V_3}{V_2} \\ &= 0. \end{aligned}$$

Note The entropy change vanishes for any reversible adiabatic process, regardless of whether the gas is monatomic or polyatomic. The logarithmic expression above assumes a constant heat capacity, but the isentropic condition itself does not require this assumption.

2.4 (7)

Since the system returns to its initial state after one complete cycle, the total change in internal energy is zero:

$$\Delta U_{\text{cycle}} = Q_{\text{net}} - W = 0.$$

Therefore,

$$W = Q_{\text{net}} = Q_{\text{in}} - Q_{\text{out}},$$

where Q_{out} denotes the magnitude of the heat released to the cold reservoir.

Since the Carnot cycle is reversible, the total entropy change over one cycle is zero:

$$\frac{Q_{\text{in}}}{T_H} - \frac{Q_{\text{out}}}{T_L} = 0.$$

With $Q_{\text{in}} = Q$, we have

$$Q_{\text{out}} = \frac{T_L}{T_H} Q.$$

Thus, the net work done by the gas, which equals the area enclosed by the cycle in the P–V diagram, is

$$\begin{aligned} W &= Q - Q_{\text{out}} \\ &= \boxed{Q\left(1 - \frac{T_L}{T_H}\right)}. \end{aligned}$$

3 III

3.1 (8)

The pressure at which the liquid and vapor phases coexist in thermal equilibrium is called the *saturated vapor pressure* (or saturation vapor pressure).

It depends only on temperature for a given pure substance in liquid–vapor equilibrium.

3.2 (9)

For the small Carnot cycle shown in the figure, the enclosed area in the P–V diagram can be approximated by

$$W \simeq (V_G - V_L)\Delta P.$$

Using the Carnot efficiency obtained in (7),

$$\begin{aligned} W &= Q\left(1 - \frac{T - \Delta T}{T}\right) \\ &= Q\frac{\Delta T}{T}. \end{aligned}$$

Therefore,

$$Q\frac{\Delta T}{T} \simeq (V_G - V_L)\Delta P.$$

Rearranging, we obtain

$$\boxed{\Delta P \simeq \frac{Q\Delta T}{T(V_G - V_L)}}.$$

In the limit $\Delta T \to 0$, this gives the Clapeyron equation:

$$\boxed{\frac{dP}{dT} = \frac{Q}{T(V_G - V_L)}},$$

where Q is the latent heat associated with the phase transition of the given amount of substance.

3.3 (10)

From the given saturation vapor pressures,

$$\begin{aligned}\Delta P &= (3.57 - 3.36) \times 10^3 \\ &= 2.1 \times 10^2 \text{ Pa},\end{aligned}$$

and

$$\Delta T = 1 \text{ K}.$$

Since $V_G \gg V_L$, we can approximate

$$V_G - V_L \simeq V_G.$$

For 1 g of water, the number of molecules is

$$N = \frac{1}{18} N_A = \frac{6.0 \times 10^{23}}{18}.$$

Assuming that the vapor behaves as an ideal gas,

$$PV_G = NkT.$$

Thus,

$$\begin{aligned}V_G &= \frac{NkT}{P} \\ &= \frac{\frac{6.0\times10^{23}}{18}(1.4 \times 10^{-23})(300)}{3.57 \times 10^3} \\ &\approx 3.9 \times 10^{-2} \text{ m}^3.\end{aligned}$$

From the result of (9),

$$Q \simeq \frac{T(V_G - V_L)\Delta P}{\Delta T} \simeq \frac{TV_G \Delta P}{\Delta T}.$$

Substituting the numerical values,

$$\begin{aligned}Q &\approx \frac{300(3.9 \times 10^{-2})(2.1 \times 10^2)}{1} \\ &\approx 2.5 \times 10^3 \text{ J}.\end{aligned}$$

Therefore, the latent heat of vaporization of 1 g of water at 300 K, to one significant figure, is

$$\boxed{Q \approx 2 \times 10^3 \text{ J}}.$$